

Validation of a balancing model based on master-slave battery management system architecture

Yu-Lin Lee^a, Chang-Hua Lin^a, Shoeb Azam Farooqui^a, Hwa-Dong Liu^{b,*}, Javed Ahmad^a

^a Department of Electrical Engineering, National Taiwan University of Science and Technology, Taipei 10607, Taiwan

^b Undergraduate Program of Vehicle and Energy Engineering, National Taiwan Normal University, Taipei 10610, Taiwan

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ABSTRACT

The battery management system (BMS) performs the monitoring and control of the charging/discharging process of the cell, state of charge estimation, battery safety and protection, state of health estimation, cell balancing, and thermal management. These control operations ensure accurate, safe, and reliable operating conditions, individual cell damage on battery string, and increase the cell life. Battery balancing is one of the key technologies in the BMS, which affects the actual usable capacity range, performance, and lifetime of the entire battery module. However, it is difficult to evaluate or compare the performance of the overall BMS under the multi-series battery module structure. Most of the proposed battery energy storage system (ESS) models focus on energy distribution and system estimation (microgrid or renewable energy). This study develops a balancing model for estimating the balancing performance of the BMS. A Master-Slave BMS (MS-BMS) is proposed to validate the balancing model. The Master and Slaves of the BMS employed a traditional flyback converter with a MOSFET switching array and bidirectional flyback converters with photoMOS arrays, respectively, to perform the balancing behavior. The balancing skill, balancing circuit power, equalization speed, control simplicity, modularization simplicity, and cost has been considered in designing the proposed BMS. The simulation results are verified with the experimental results, which are then compared with the balancing model's results under the five balancing conditions, and the average absolute error of the balancing duration results is 3.12%. The performance of the proposed BMS is also compared with the existing modular BMS architecture, which validates the advantages of the proposed BMS including equalization speed, control simplicity, modularization simplicity, and cost.

1. Introduction

An energy storage system (ESS) is a competent alternative to the fossil fuel-based energy system. It plays a paramount role in the adoption of renewable energy to combat climate change issues and its integration into the grid. ESS may comprise various supercapacitors or batteries and are extensively used in electric vehicles (EVs), off-grid solar microgrids, and portable electric devices. In supercapacitors, energy is stored in the form of static charges. While in the battery cells, chemical energy changes into electrical energy or vice versa. Depending on the application, the user can select a fuel cell, ultra-capacitors, super-capacitors, or batteries for the ESS. To store energy in transportation and renewable energy systems, high-specific energy and the power-storage system are needed. The battery is a good choice for the storage system as it has an appropriate specific power, high specific energy density, and a long rate of discharge. Nickel-cadmium, lead-acid, zinc-air, nickel-metal hybrid,

and lithium-ion batteries are among the several forms of these batteries. However, a single cell has finite voltage irrespective of the type of cell material, where the voltages of Ni-Cd, lead-acid, and Li-ion batteries are 1.2 V, 2.0 V, and 3.7 V, respectively. Different applications have dissimilar specifications of battery power systems as a result of the various scenarios. Even though all batteries are employed in many applications, lithium-ion batteries have obtained increasing popularity in many applications, such as ESS, EVs, and uninterruptible power supplies due to their high energy density, lower weight, low self-discharge, higher safety and longer lifespan. The cells are connected in series, parallel, or a combination of both to get the higher power and desired voltage for achieving the required specification of lithium-ion battery power. For efficient operation of the connected pack of cells, the voltage and capacity of each connected cell should be compatible. The voltage and capacity of each cell are limited and the cell undergoes the charging-discharging process during operation. After numerous charging-

* Corresponding author.

E-mail address: hdliu@ntnu.edu.tw (H.-D. Liu).

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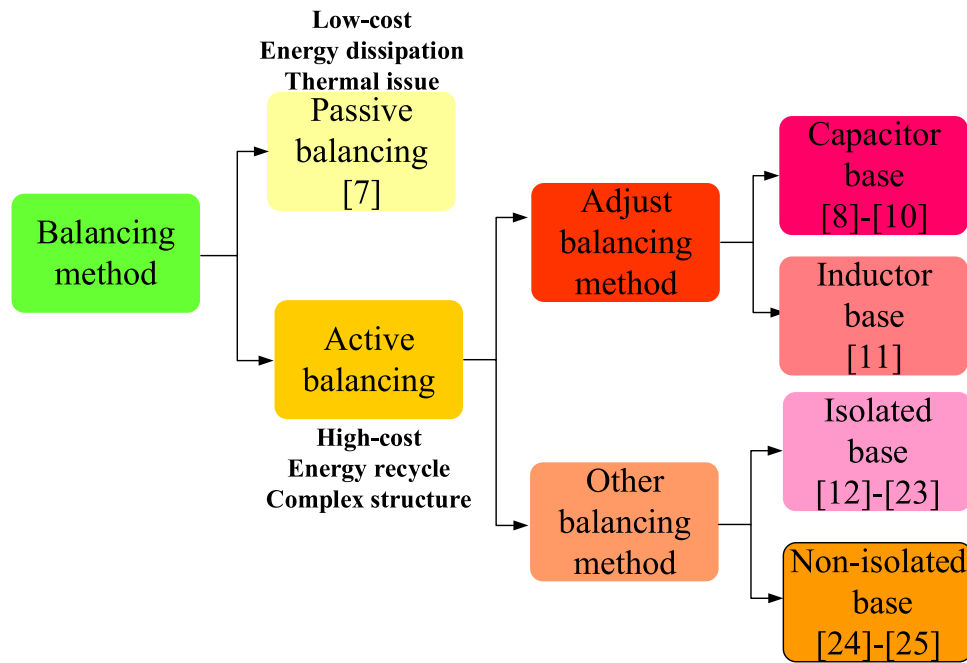


Fig. 1. Categories of the balancing methods.

discharging cycles, the energy storage capability of the cell slowly decreases. This leads to the cell mismatch problem because of the cell's different chemical and electrical characteristics. In the long run, this problem may impact the state of charge (SOC)/ state of health (SOH) characteristics of the battery power system and increase the operational risk of the battery power system. Hence, the development of a BMS, which provides monitoring, protection, and cell balancing functions to the battery system, is considerable [1–6]. A thermal battery module thermal management system is presented as a vital part of the electric vehicle to cool the batteries using the air conditioning refrigerant [1–2] or liquid refrigerant [3]. Integrating renewable energy sources with the installation of the battery ESS into the isolated power grid also shows the necessity of the BMS [4]. In addition, the BMS for similar integration, known as vehicle-to-building interactions, is another critical application that can also be improved by novel energy management strategies [5]. Furthermore, the BMS can be necessarily adopted in hybrid powertrain technology (i.e., batteries and supercapacitors), improving the performance of the power distribution algorithm [6].

Several topologies and methods have been proposed in the literature for cell balancing in the BMS [7–34]. Releasing redundant energy, transferring energy between cells, or controlling the charging/discharging energy of individual cells are certain approaches that are usually used in these balancing techniques. Based on energy balancing techniques and energy storage elements, these balancing methods can be categorized as passive balancing and active balancing. Passive balancing is commonly used due to its lower cost and simple circuitry. In this method, the excess charge or energy of the fully charged cells is removed through a resistor till the energy of the cell reaches the reference mark. This passive element i.e. resistor can be deployed in fixed or switched mode. However, these circuits suffer from low efficiency and energy loss. Moreover, the thermal dissipation problem exists in the passive balancing method, which must be considered to ensure the safe operation of the battery system [7]. These shortcomings can be overcome by implementing active balancing methods. These methods present higher efficiency as there is no energy dissipation in the circuit but the transfer of excessive energy or charge from a higher energy cell to a lower energy cell. This transfer could be cell to cell, cell to module, module to module, or module to pack. These methods have high balancing speed and high efficiency. Capacitors [8–10], inductors [11],

or isolated/non-isolated converters [12–25] can be used for transferring the energy from the most charged cell to the less charged cell or redistribute the energy among the cells and help them in the balancing. Capacitor based could be switched or modularized, inductor based could be single or multiple, transformer based may be single or multiple winding and the converter based could be a boost, buck-boost, flyback, Cuk, full bridge, or resonant. Though the circuits employing the inductor or transformer as the active element have high efficiency and balancing speed but they are bulky and suffer from magnetic loss. While the converter based circuits are bidirectional, have low current and voltage stress, and have higher efficiency and balancing speed. The number of components in these circuits are extremely high which makes its design more complex and added to its cost as well. A general classification of the balancing methods has been depicted in Fig. 1.

Battery systems are preferably designed for higher voltages such as 400 V, 760 V, or 1000 V to minimize the power loss during the power transmission. Cells are modularized to have a safe and reliable BMS for a high voltage battery power system including the monitoring, protection and balancing of the cells. However, cell imbalance is a serious issue in the battery system design which must be handled carefully otherwise it may cause severe effects on battery health and safety. Cell modularization is an effective method to get rid of the imbalance issue as well as minimize the components [26–28]. In [26], a modularized automatic equalizer using multi-winding transformers is proposed for battery cell balancing. The proposed architecture can transfer the energy of the cell having the most power to other cells in the whole pack. Nevertheless, this method had quite difficult controls of the balance current and the power flow. In [27], a modularization method was proposed by Hamed et al., where the series cells were divided into several modules, and the balancing function was realized via a multioutput transformer and multiplexer switches. However, the balancing efficiency was low as the method could only be responsible for balancing a single cell in each battery module at the same time. In [28], Lee et al. designed a modularized design of an active cell-to-cell/module-to-module charge equalizer with a charge equalization algorithm for a Li-ion battery pack. The active cell-to-cell and module-to-module charge equalizers had the same circuit topology with different specifications, using four bidirectional switches and an inductor as an energy carrier. The architecture is simple and the control of the balancing process is easy, but the cost is

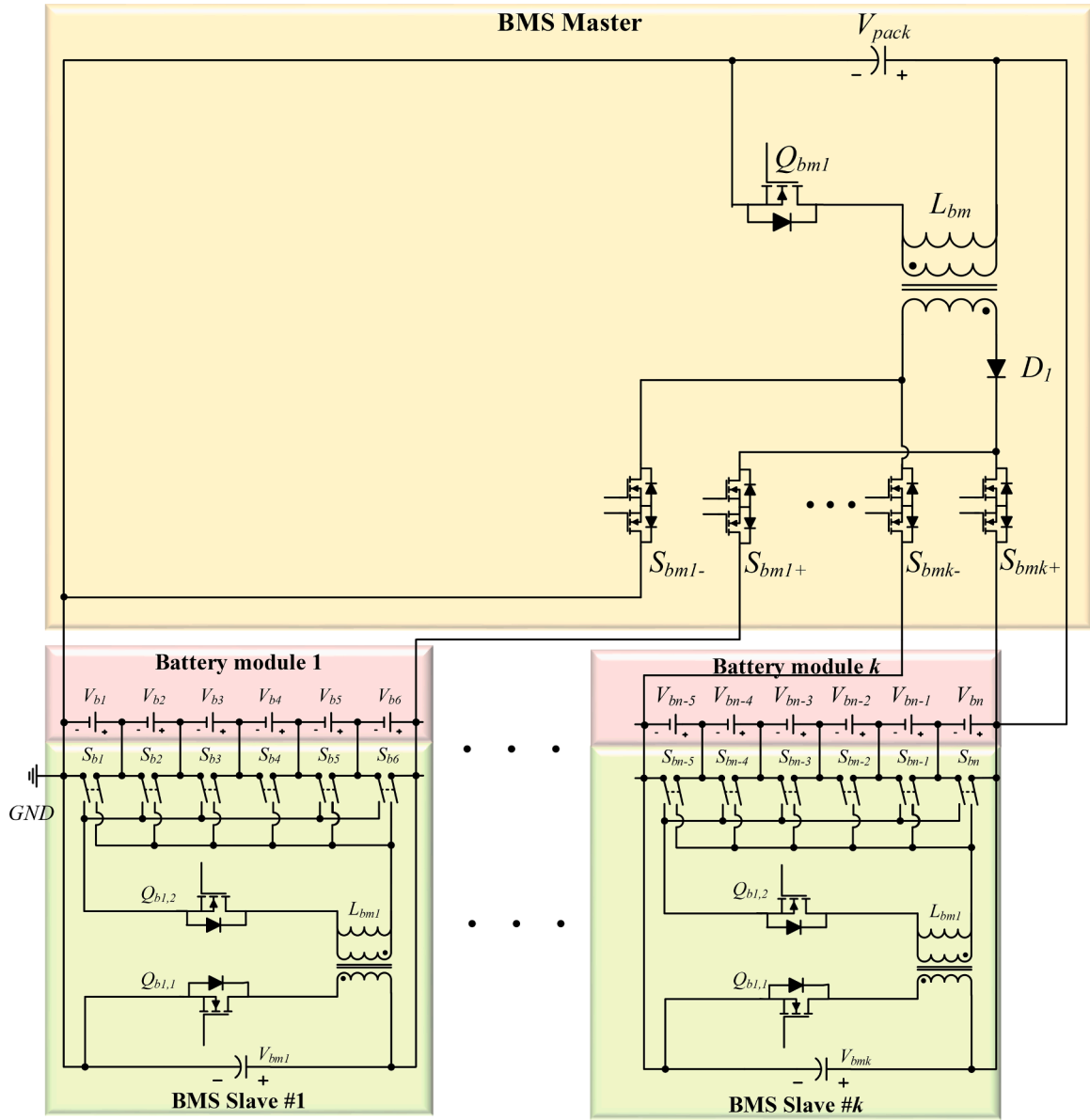


Fig. 2. Basic circuit structure of the proposed MS-BMS.

quite high and the non-isolated converter is dangerous in high voltage BMS.

A few more models of the battery ESS (BESS) are presented [29–33]. Delnia Sadeghi et al. proposed two algorithms for residential buildings, namely the Multi-Objective Particle Swarm Optimization, and the Non-Dominant Sorting Genetic Algorithm II, to minimize the life cycle cost, probability of loss of power supply, and CO₂ emissions [29]. Hadi Niaei et al. presented a peer-to-peer-based transactive energy sharing architecture for the ESS as two-stage mixed integer non-linear programming, using smart demand response programming integrated with a machine learning approach, i.e., radial basis neural network [30]. Seyed Ehsan et al. presented a novel decentralized bi-level stochastic optimization approach based on the progressive hedging algorithm for multi-agent systems in multi-energy microgrids to enhance network flexibility [31]. Hamid Reza et al. introduced a smart charging approach for off-board EV chargers in Home-Energy-Hub applications along with DC sources such as Photovoltaic (PV) and Battery Storage (BS), especially in residential areas [32]. Hamid et al. also implemented a system with an accelerated integration of renewable energy resources and dispersed generations, contributing to big shifts in the power grid [33].

This study focuses on the balancing process of the BMS. This study develops a balancing model based on the proposed MS-BMS architecture to evaluate the balancing performance before implementing the system. The balancing process can be evaluated under different voltage distribution conditions.

To find out a compromise among balancing skill, balancing circuit power, equalization speed, control simplicity, modularization simplicity, and cost, an MS-BMS architecture balancing model for a high-count series battery power system has been proposed in this paper. The proposed BMS design considers operational reliability, system complexity, sampling accuracy, and balancing duration. The basic circuit structure consists of battery modules, BMS Slaves, and a BMS Master. The battery modules are formed by the series connection of several cells. The BMS Slaves provide the balancing functions for each battery module, while the BMS Master is designed to solve the imbalance problem among the battery modules. Modularized BMS measures various parameters, including the current, voltage, and temperature. The control core collects these data through the Master-Slave Modbus communication protocol and stores them in some data storage system, e. g., SD card, to ensure the safe operation of the proposed system.

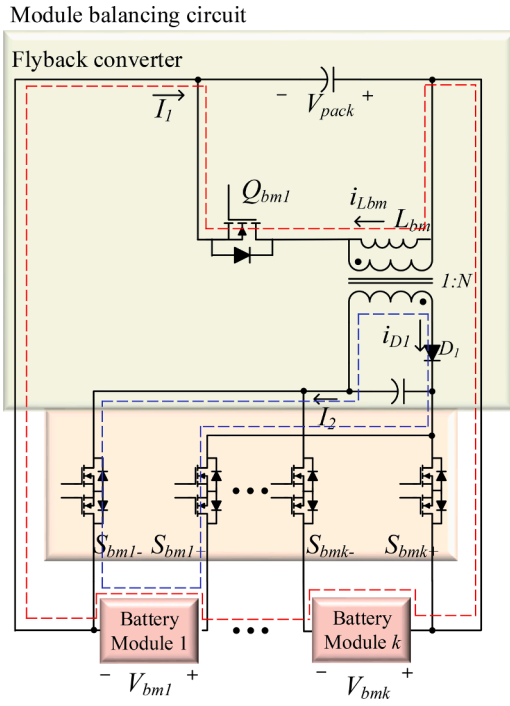


Fig. 3. Schematic of the proposed module balancing circuit.

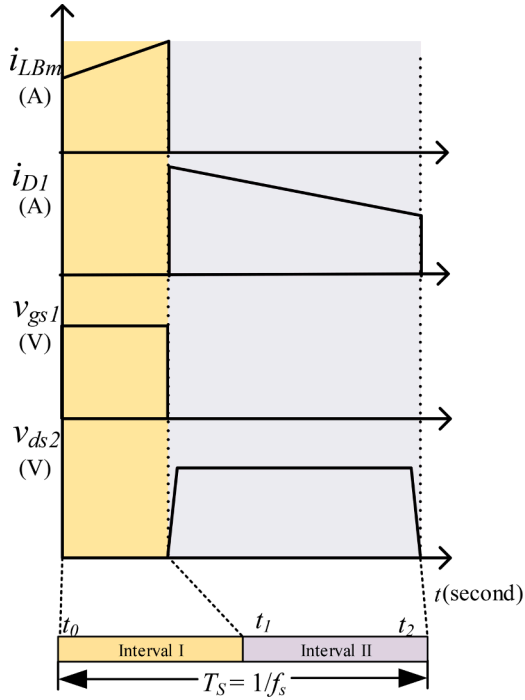


Fig. 4. Waveforms of the proposed balancing circuit.

Compared with the passive balancing method, both balancing methods implemented in the MS-BMS architecture can recycle the energy dissipation caused by the passive balancing method [7]. Other active balancing methods, such as capacitor-based [8–10], inductor-based [11], isolated-based [12–23], and unisolated-based balancing methods [24–25], the implemented flyback converter based balancing method with switching array has better performance due to operational reliability, system complexity, sampling accuracy, and balancing duration. Additionally, in most BMS research, the architecture's performance can

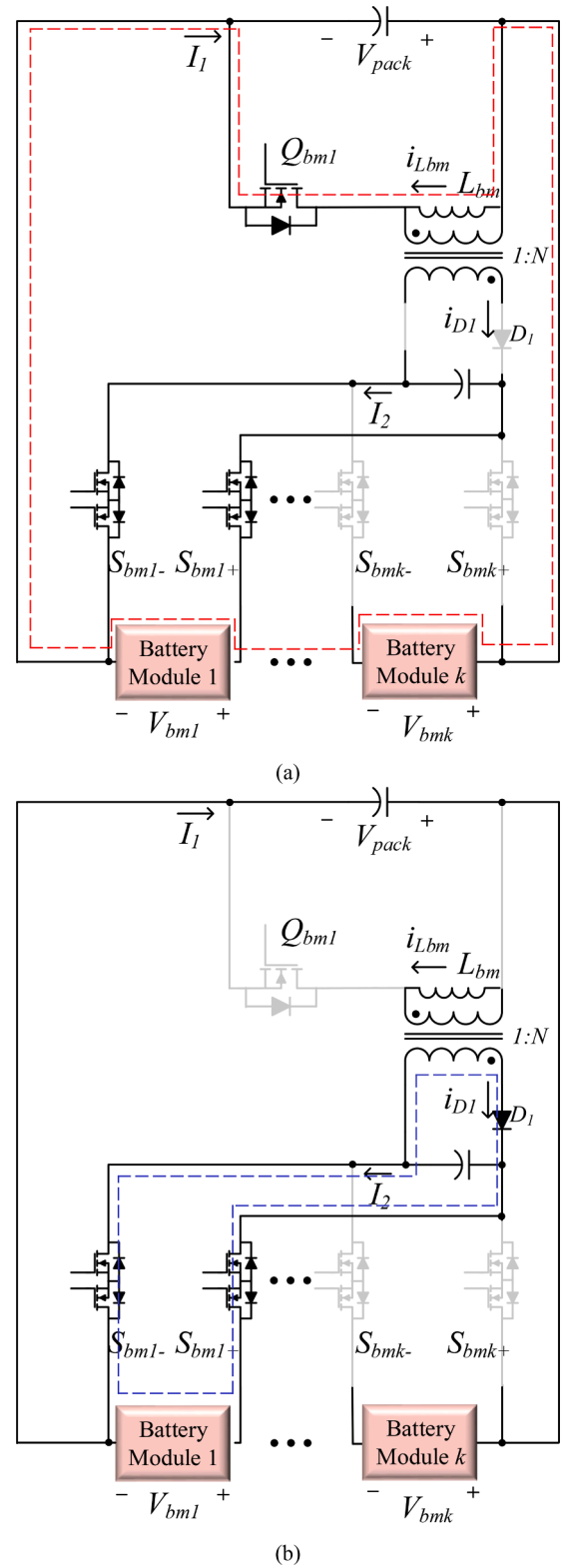


Fig. 5. Operational mode of the proposed balancing circuit (a) Mode I: battery pack is discharging (b) Mode II: battery module 1 is charging.

only be obtained after implementation. However, this study also raises a balancing model, which considers lithium ternary cell's mathematical model, and the operating principle of the BMS Master and the BMS Slaves. The proposed balancing model can estimate the balancing performance before implementation. The proposed balancing model has been simulated in the Matlab/Simulink model. The corresponding

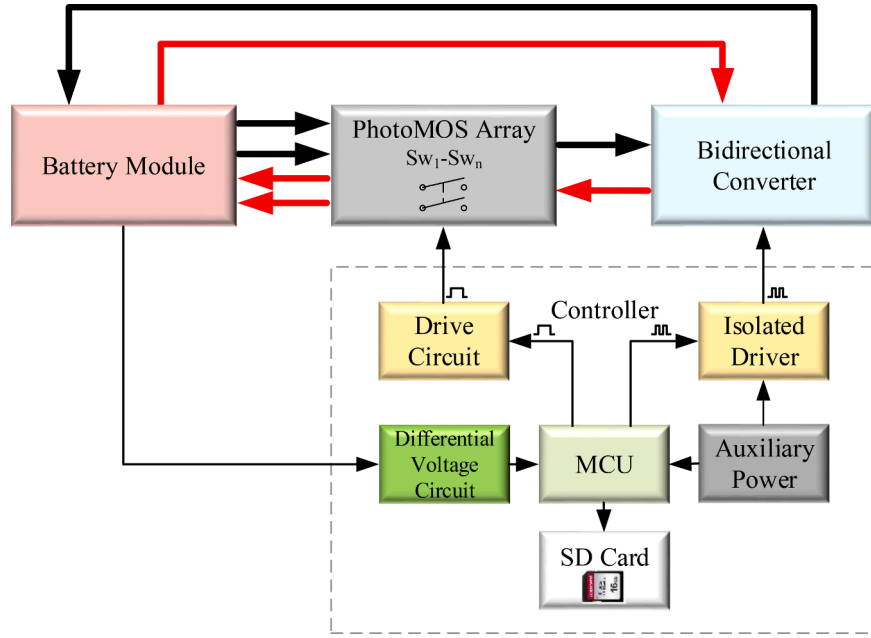


Fig. 6. Block diagram of the modularized BMS based on a bidirectional converter.

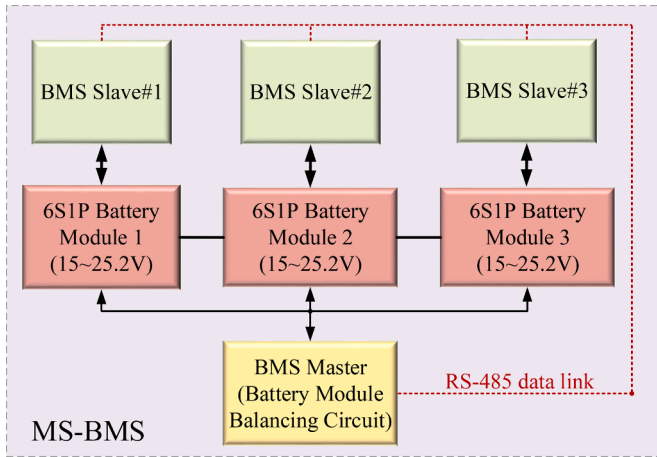


Fig. 7. Block diagram of the proposed MS-BMS with Modbus protocol.

hardware system has been implemented to verify its feasibility and performance. Moreover, the simulation results have been validated on the experimental prototype under five different conditions that helps to evaluate the balancing efficiency of the proposed BMS architecture.

The remaining work has been structured as follows: Section 2 presents the system descriptions of the proposed MS-BMS balancing system which includes modular BMS design, BMS Master, BMS Slaves and MS-BMS architecture. Section 3 describes the proposed MS-BMS balancing model. In Section 4, both the simulation and hardware results are discussed, which verify the feasibility and performance of the proposed system. Lastly, the paper is concluded in Section 5.

2. System descriptions of the proposed MS-BMS balancing system

Fig. 2 shows the basic circuit structure of the MS-BMS design, including battery modules, a BMS Master, and BMS slaves. In the proposed BMS architecture, the cells are connected in series to form a battery module. The BMS Slaves can provide balancing functions for each cell in the battery module while the BMS Master is designed to

solve the imbalance problem among the battery modules. Through the proposed BMS, the cells can reach equilibrium with several periods of energy transmission.

2.1. BMS master

BMS Master is designed to solve the imbalance problem among the battery modules. The proposed balancing circuits consist of a flyback converter and a switching array which is shown in Fig. 3. Several battery modules are connected in series to form a battery pack. Here, k number of battery modules are used to form a battery pack. The battery pack voltage is applied at the input of the flyback converter while the output voltage terminals of the flyback converter are connected to the switching array. The transformer employed in the flyback converter has the turn ratio of N and the magnetizing inductance of L_{bm} and i_{Lbm} is the current flowing through it. D_1 works as a rectifier diode and i_{D1} is the current flowing through it. A MOSFET Q_{bm1} is used for power switching. S_{bmk} is the switching array; I_1 is the charge flow rate from the battery pack whereas I_2 is the module balancing current. The switching frequency of the proposed balancing circuit in continuous conduction mode (CCM) is $f_s = 1/T_s$ which is mentioned in the theoretical waveform depicted in Fig. 4. Let's take a condition to explain the operating principles of the proposed balancing circuit in which the voltage of the 1st battery module V_{bm1} is the lowest among the battery modules. In this case, Q_{bm1} controls the circuit as the main switch and S_{bm1+} as well as S_{bm1-} will be turned ON. The proposed balancing circuit operates in two distinct modes which are demonstrated Fig. 5.

Mode I (t_0-t_1): In this mode, MOSFET Q_{bm1} is ON. The supply is provided by V_{pack} and the energy is stored in the L_{bm} . This is the discharging mode of the battery pack. The current i_{Lbm} increases linearly and can be expressed as Eq. (1). At time $t = t_1$, MOSFET Q_{bm1} is turned OFF.

$$i_{Lbm}(t) = i_{Lbm}(t_0) + \frac{V_{pack}(t - t_0)}{L_{bm}} \quad (1)$$

Mode II (t_1-t_2): In this mode, MOSFET Q_{bm1} is OFF and D_1 is ON. The energy stored in L_{bm} is transferred to the V_{bm1} . This is the charging mode of the battery pack. The current i_{Lbm} decreases linearly and can be expressed as Eq. (2). At time $t = t_2$, MOSFET Q_{bm1} is turned ON.

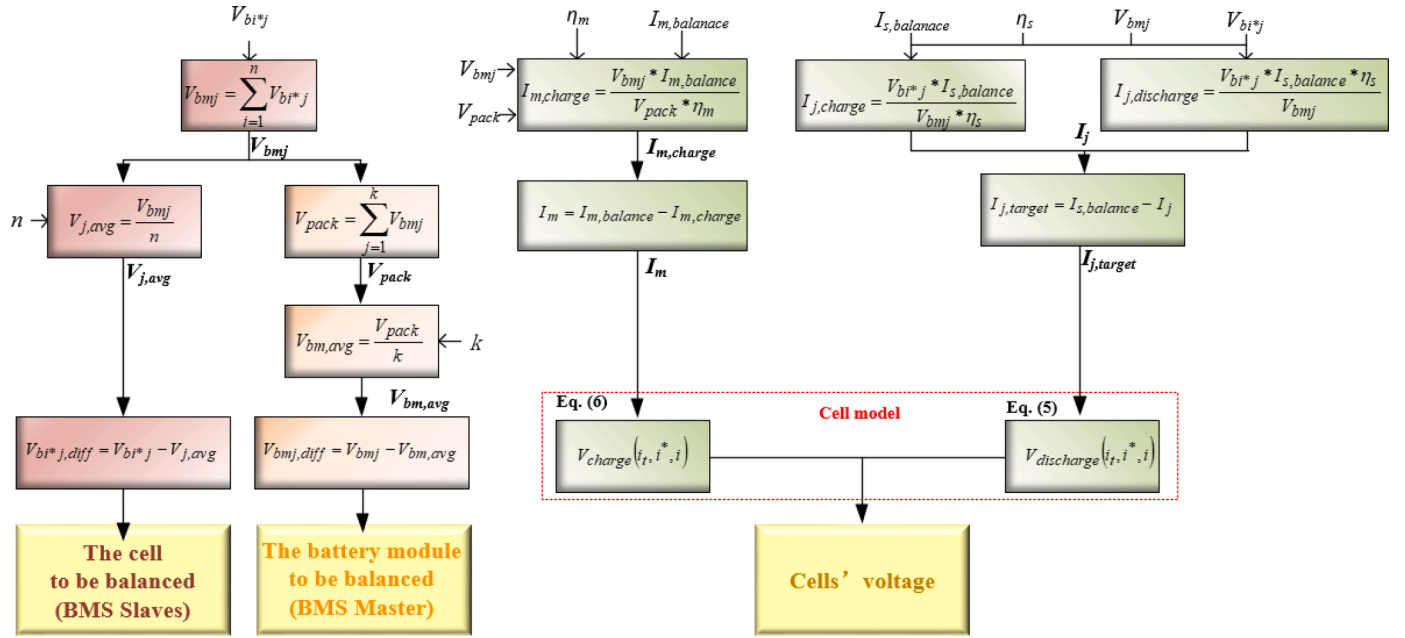


Fig. 8. Block diagram of the proposed mathematical balancing model.

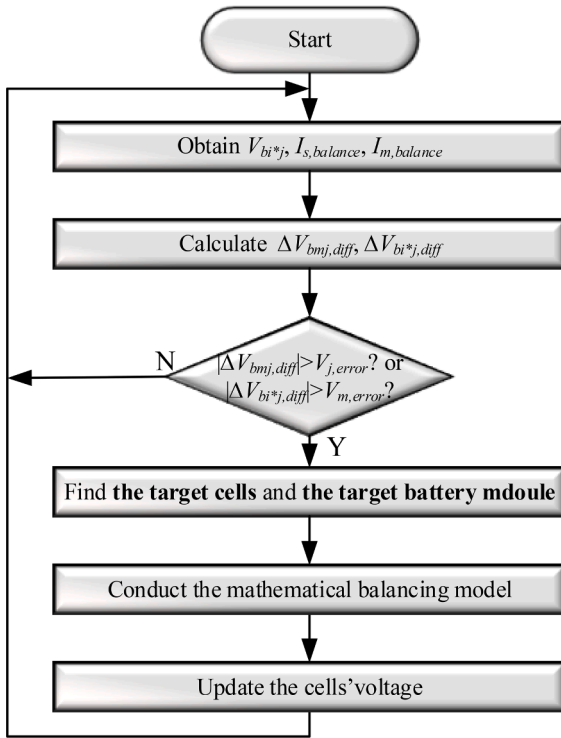


Fig. 9. Flowchart of the balancing strategy for the proposed MS-BMS.

$$M = \frac{V_o}{V_g} = N \frac{D}{1-D} \quad (4)$$

2.2. BMS slaves

The BMS Slaves used in the proposed system are based on the research conducted in [34], where all the monitoring, active cell balancing, and protection functions are implemented for 6S1P battery modules. A bidirectional converter based active cell balancing system has been used here which is shown in Fig. 6. During the balancing process, the energy is transferred within the battery module through the PhotoMOS array and the bidirectional converter. An offline data storage device i.e. SD card is deployed to save the data of every cell such as voltage and current and helps to prevent data loss during the power cut or communication failure. Along with the simple architecture, bidirectional energy transfer, and being inexpensive, the modularized BMS slaves can achieve high balancing efficiency. Thus, plays a crucial role in the MS-BMS architecture.

2.3. Master-Slave BMS architecture

Modbus communication structure is the source of inspiration for the proposed Master-Slave BMS architecture design. The higher voltage requirement of the BESS such as 400 V, 800 V, or even higher can easily be fulfilled with the proposed BMS architecture. Let's consider a battery power system with a 18S1P battery pack to verify the feasibility of the proposed architecture. Fig. 7 displays the block diagram of the proposed BMS with Modbus protocol. The BMS consists of a BMS Master, three BMS Slaves and three 6S1P battery modules. Primarily, the BMS Slaves provide the active cell balancing for each 6S1P battery module. Secondly, BMS Master helps to solve the imbalance problem among the three 6S1P battery modules. Modularized BMS measures the various parameters including the current, voltage and temperature. These data are collected by the control core through the Master-Slave Modbus communication protocol to ensure the safe operation of the proposed system. Furthermore, the data of each cell are stored in some data storage system e.g. SD card. Thus, the data are saved and can further be used to estimate the status of the cell by training the neural networks using these data [35–37] and can be applied in many more applications.

$$i_{Lbm2}(t) = i_{Lbm}(t_1) - \frac{V_{bm1}}{L_{bm}}(t - t_1) \quad (2)$$

From Eqs. (1), (2), and volt-second balance theory, the voltage gain M of the flyback converter in the CCM is obtained which is expressed in Eqs. (3) and (4).

$$\int_0^{T_s} L_{bm} i_{Lbm}(t) dt = \int_{t_0}^{t_1} V_{pack} dt + \int_{t_1}^{t_2} \left(-\frac{V_{bm1}}{N} \right) dt = 0 \quad (3)$$

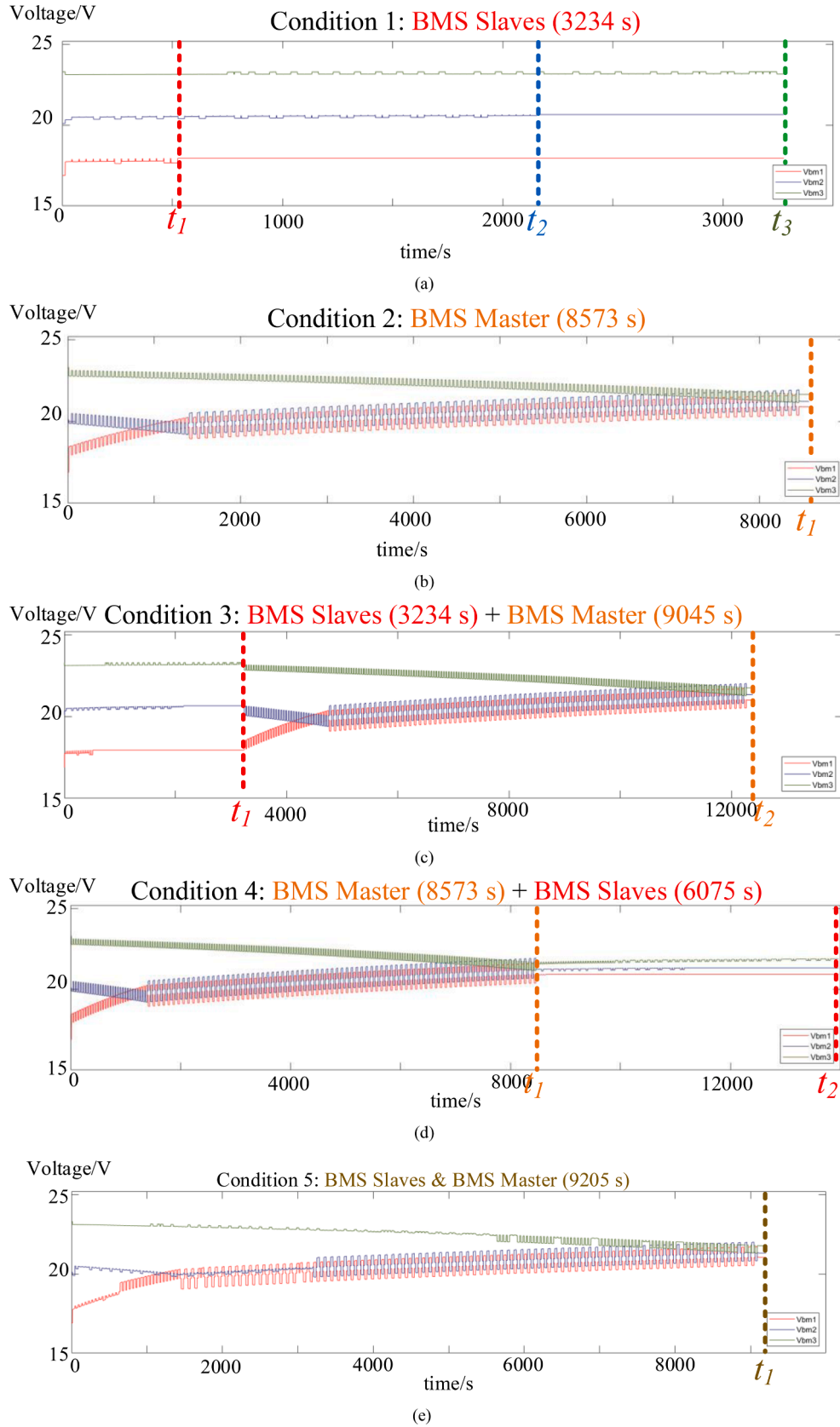
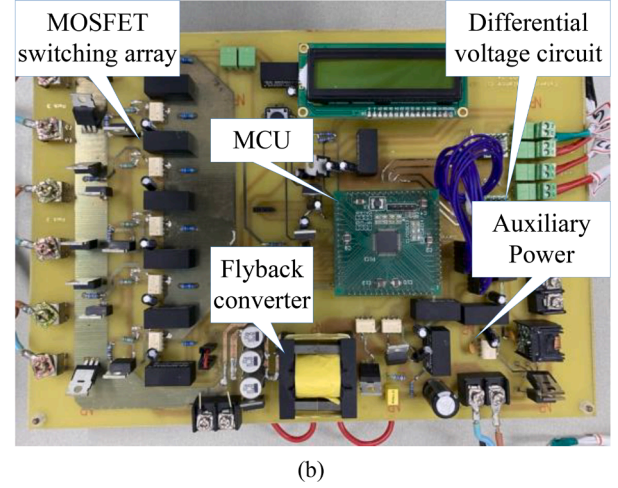
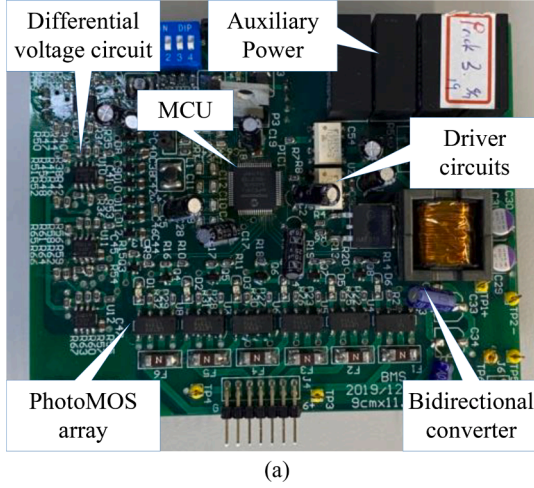
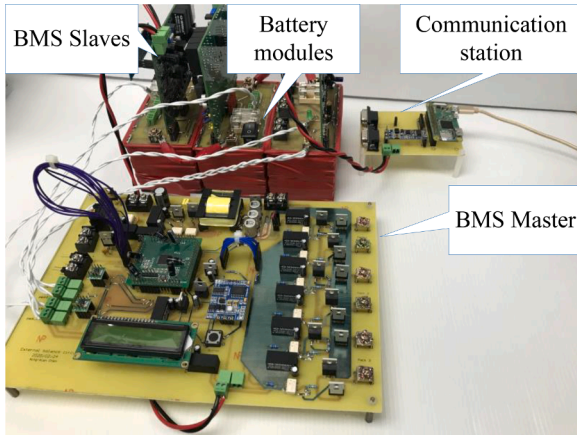


Fig. 10. The simulation results of the proposed BMS mathematical model: (a) Condition 1: BMS Slaves operates, (b) Condition 2: BMS Master operates, (c) Condition 3: BMS Slaves operates + BMS Master operates, (d) Condition 4: BMS Master operates + BMS Slaves operates, and (e) Condition 5: BMS Master & Slaves operate simultaneously.

Table 1

The simulation results of the proposed BMS under various conditions.

Condition	Step 1 BMS Status	Duration	$\Delta V_{bm,max}$	$\Delta V_{b,max}$	$\Delta V_{b,diff,max}$	Step 2 BMS Status	Duration	$\Delta V_{bm,max}$	$\Delta V_{b,max}$	$V_{b,diff,max}$
1	Slaves	3234 s	4.61 V	0.82 V	0.03 V	/	/	/	/	/
2	Master	8573 s	0.15 V	0.55 V	0.35 V	/	/	/	/	/
3	Slaves	3234 s	4.61 V	0.82 V	0.03 V	Master	9045 s	0.15 V	0.14 V	0.07 V
4	Master	8573 s	0.15 V	0.55 V	0.35 V	Slaves	6075 s	0.37 V	0.11 V	0.03 V
5	Master Slaves	9205 s	0.15 V	0.06 V	0.03 V	/	/	/	/	/

**Fig. 11.** Prototype of the proposed BMS: (a) BMS Slave and (b) BMS Master.**Fig. 12.** Complete experimental setup of the proposed MS-BMS.

It can be concluded that each BMS Slave provides an active cell balancing function for a single 6S1P battery module, and the BMS Master overcomes the imbalance phenomenon among three 6S1P battery modules. Thus, the proposed BMS can reach the 18S1P BESS equilibrium easily and safely through the Modbus communication protocol.

3. Balancing model based on the proposed MS-BMS architecture

To verify the feasibility of the proposed MS-BMS architecture, a mathematical balancing model has been presented in this study. A batch of lithium ternary cells (UR18650NSX, Panasonic) with a rated capacity of 2600 mAh and a voltage range of 2.5 V – 4.2 V for a single cell has been used in the battery system. A lithium ternary cell UR18650NSX model is created in Matlab (ver. 2020a) [38], whose equations can be

Table 2

Specification of the components used in the proposed MS-BMS design.

Components/Parameters			Specifications
Balancing circuit	BMS Master	MOSFET switch	SUP90330E(Vishay Siliconix)
		MOSFET (Switching array)	FB52N15D(Inchange SEMI Company)
		Transformer	Turn Ratio 2:3
			$L_{m,m}$ 40 μ H
		Balancing current $I_{m,balance}$	1 A
	BMS Slaves	MOSFET switch (low side)	CSD18510KCS(Texas Instruments)
		MOSFET switch (high side)	CSD19536KTT(Texas Instruments)
		PhotoMOS array	LT318(Letex TECH. Corp.)
		Transformer	Turn Ratio 1:3
			$L_{m,s}$ 117 μ H
Lithium-ion battery		Balancing current $I_{s,balance}$	1 A
		Part number	UR18650NSX (Panasonic)
		Capacity	2600 mAh
		Voltage range	Cell 2.5 V–4.2 V Module 15 V–25.2 V Pack 45 V–75.6 V

obtained as follows:

$$V_{discharge}(i_t, i^*, i) = E_0 - K \cdot i^* \cdot \frac{Q}{Q - i_t} - K \cdot i_t \cdot \frac{Q}{Q - i_t} + A \cdot \exp(-B \cdot i_t) \quad (5)$$

$$V_{charge}(i_t, i^*, i) = E_0 - K \cdot i^* \cdot \frac{Q}{i_t + 0.1 \cdot Q} - K \cdot i_t \cdot \frac{Q}{Q - i_t} + A \cdot \exp(-B \cdot i_t) \quad (6)$$

where i_t – Extracted capacity (Ah), i^* – Low-frequency current dynamics (A), i – Battery current (A), E_0 – Constant voltage (V), K –

Table 3

Cell voltages in the unbalanced 18S1P battery pack under various testing conditions.

Module1	V_{b1}	V_{b2}	V_{b3}	V_{b4}	V_{b5}	V_{b6}	V_{bm1}
Voltage(V)	2.71	2.74	2.75	2.86	2.94	3.02	17.02
Module2	V_{b7}	V_{b8}	V_{b9}	V_{b10}	V_{b11}	V_{b12}	V_{bm2}
Voltage(V)	3.12	3.17	3.28	3.41	3.5	3.58	20.06
Module3	V_{b13}	V_{b14}	V_{b15}	V_{b16}	V_{b17}	V_{b18}	V_{bm3}
Voltage(V)	3.74	3.86	3.92	3.96	3.97	3.99	23.39

Table 4

Differences among the five balancing conditions.

Condition	1	2	3	4	5
Step 1	S	M	M	S	M&S (simultaneously)
Step 2			S	M	

S–BMS Slaves operates; M–BMS Master operates.

Polarization resistance (Ω), Q – Maximum battery capacity (Ah), A – Exponential voltage (V), B – Exponential capacity (Ah^{-1}).

Numerous researches have been done about battery cell modeling. The equivalent circuit model with the combination of capacitance and resistance is the most common model [39–41]. However, SOC is considered as the main variable for representing the voltage behavior of the cell model used in this study. Since every proposed model has some imperfection. The presented model gives an inaccurate result with the varying current while working satisfactorily under the constant current. So, a constant balancing current has been considered in the proposed BMS to get an accurate result. MS–BMS balancing model has been proposed in this paper. The BMS Slaves provide the balancing in each cell of the battery module and BMS Master provides the balancing of each module in the battery pack.

For BMS Slaves, the balancing cell (target) has two different modes: one (over-discharged) is being charged while the other (overcharged) is discharging. And then the formulas of the current for the other cell are shown as follows:

$$I_{j,\text{discharge}} = \frac{V_{bi*j} * I_{s,\text{balance}} * \eta_s}{V_{bmj}} \quad (7)$$

$$I_{j,\text{charge}} = \frac{V_{bi*j} * I_{s,\text{balance}}}{V_{bmj} * \eta_s} \quad (8)$$

where $I_{j,\text{discharge}}$ is the current of the target cell during discharging mode, V_{bi*j} is the voltage of the i^{th} cell in the j^{th} module, V_{bmj} is the voltage of the j^{th} module, $I_{s,\text{balance}}$ is the balancing current of the BMS Slave, $I_{j,\text{charge}}$ is the current of the target cell during charging mode, η_s is the conversion efficiency of the bidirectional flyback converter in the BMS Slave.

During the operation of the BMS Slaves for balancing the target cells, the current of the target cells in the corresponding module can be expressed as follows:

$$I_{j,\text{target}} = I_{s,\text{balance}} - I_j \quad (9)$$

where $I_{j,\text{target}}$ is the current for the target cell in the BMS Slaves, I_j is the current of the other cells. I_j is $I_{j,\text{charge}}$ during charging status, while I_j is $I_{j,\text{discharge}}$ during discharging status.

For BMS Master, the balancing module (target) can only be charged. So, the formula of the current for the other module is shown as follows:

$$I_{m,\text{charge}} = \frac{V_{bmj} * I_{m,\text{balance}}}{V_{\text{pack}} * \eta_m} \quad (10)$$

where V_{bmj} is the voltage of the j^{th} module, $I_{m,\text{balance}}$ is the balancing current of the BMS Master, V_{pack} is the voltage of the battery pack, η_m is the conversion efficiency of the traditional flyback converter in the BMS Master.

During the operation of the BMS Master for balancing the target battery module, the current of the target module can be expressed as follows:

$$I_m = I_{m,\text{balance}} - I_{m,\text{charge}} \quad (11)$$

The voltage of cells (V_{bi*j}) can be calculated by substituting $I_{j,\text{target}}$ and I_m into (5) and (6). Many other variables are also required before deciding the target cells and the target battery module. These variables are the voltage of battery modules (V_{bmj}), the voltage of the battery pack (V_{pack}), the average cell voltage of j^{th} battery module ($V_{j,\text{avg}}$), the average voltage of the battery module ($V_{bm,\text{avg}}$), the voltage difference between V_{bi*j} and $V_{j,\text{avg}}$ ($V_{bi*j,\text{diff}}$), and the voltage difference between V_{bmj} and $V_{bm,\text{avg}}$ ($V_{bmj,\text{diff}}$). The corresponding formulas are presented as follows:

$$V_{bmj} = \sum_{i=1}^n V_{bi*j} \quad (12)$$

$$V_{\text{pack}} = \sum_{j=1}^k V_{bmj} \quad (13)$$

$$V_{j,\text{avg}} = \frac{V_{bmj}}{n} \quad (14)$$

$$V_{bm,\text{avg}} = \frac{V_{\text{pack}}}{k} \quad (15)$$

$$V_{bi*j,\text{diff}} = V_{bi*j} - V_{j,\text{avg}} \quad (16)$$

$$V_{bmj,\text{diff}} = V_{bmj} - V_{bm,\text{avg}} \quad (17)$$

Fig. 8 illustrates the block diagram of the proposed mathematical balancing model, where the inputs of the proposed mathematical balancing model are cell voltage V_{bi*j} , the balancing current of the BMS Slaves $I_{s,\text{balance}}$, the balancing current of the BMS Master $I_{m,\text{balance}}$, the efficiency of the BMS Slaves η_s , the efficiency of the BMS Master η_m , the number of the cells n , and the number of the battery module k . One function of the proposed mathematical balancing model is to update the voltage of the cells by (5) and (6). The other functions are to decide the target cells (BMS Slaves) and the target battery module (BMS Master) to be balanced. Fig. 9 is the flowchart of the proposed mathematical balancing model, where $V_{j,\text{error}}$ is the preset voltage as the balancing start standard of the BMS Slaves, $V_{m,\text{error}}$ is the preset voltage as the balancing start standard of the BMS Master. The designed mathematical balancing model has been implemented in a simulation model. Furthermore, the simulation results have been validated on the experimental prototype to verify the feasibility and performance of the proposed BMS architecture.

4. Simulation results and hardware validation of the proposed model

4.1. Simulation results

The proposed balancing model has been implemented and simulated in the Matlab/Simulink model. The performance of the proposed BMS architecture can be seen from the simulation results. Fig. 10 demonstrates the simulation results of the proposed model presenting the variation of cell voltage (V_{bn}) with respect to the time (t). $\Delta V_{bm,\text{max}}$ is the difference between the highest and the lowest voltage of the battery module, $\Delta V_{b,\text{max}}$ is the difference between the highest and the lowest voltage of the cell and $V_{b,\text{diff},\text{max}}$ is the maximum value of the $V_{bi*j,\text{diff}}$ [Eq. (16)].

For Condition 1, the process took 3234 s to attain the balanced condition which is shown in Fig. 10(a). Where, $\Delta V_{bm,\text{max}} = 4.61$ V, $\Delta V_{b,\text{max}} = 0.82$ V, and $V_{b,\text{diff},\text{max}} = 0.03$ V. During the balancing process, BMS Slave #1 achieve the balanced condition for battery module 1 at $t = t_1$, BMS Slave #2 achieve the balanced condition for battery module 2 at $t = t_2$ and the BMS Slave #3 achieve the balanced condition for battery module 3 at $t = t_3$. For Condition 2, the process took 8573 s to attain the balanced condition which is shown in Fig. 10(b). Where, $\Delta V_{bm,\text{max}} =$

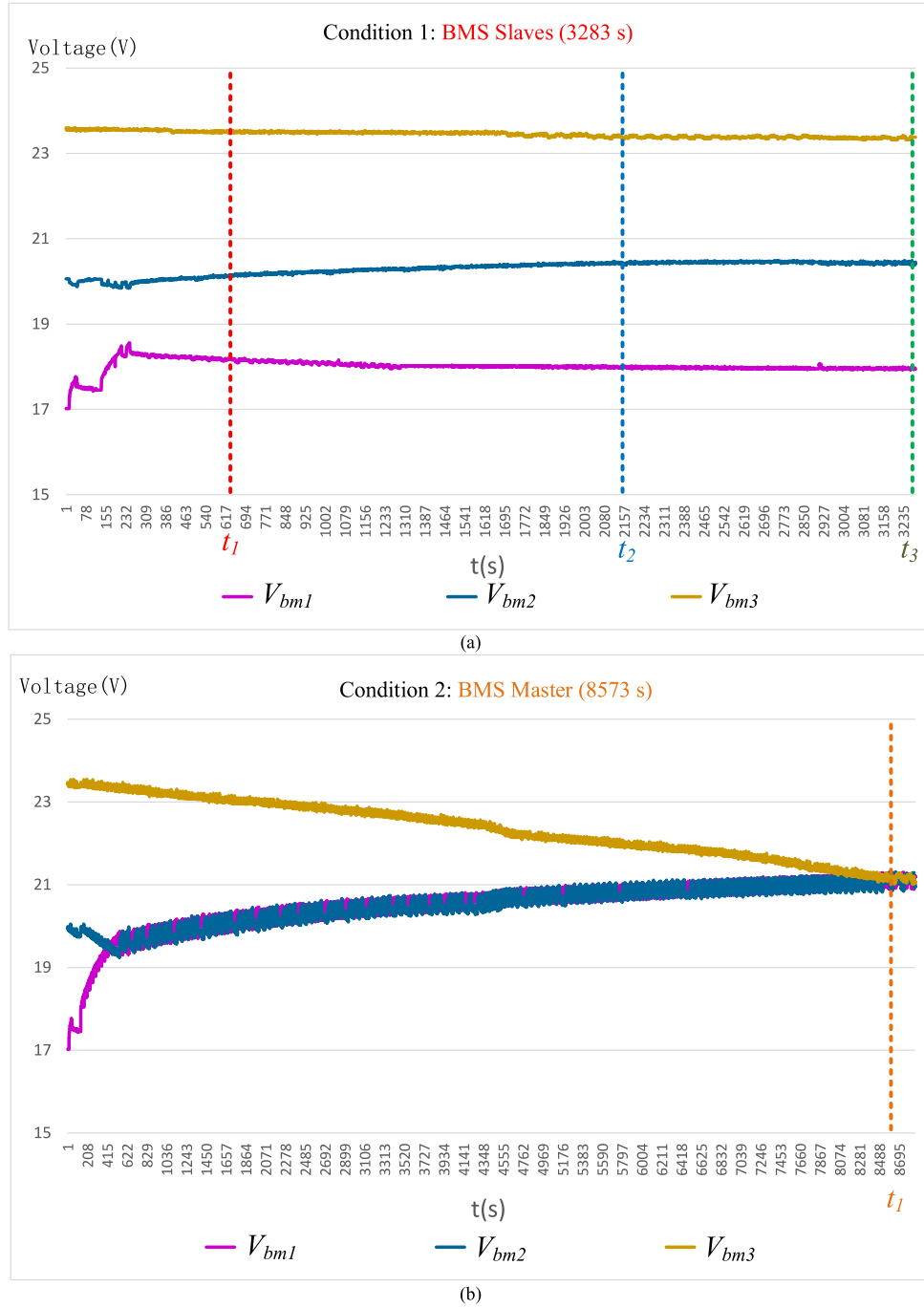


Fig. 13. The hardware results of the proposed BMS under different conditions: (a) Condition 1: BMS Slaves operates, (b) Condition 2: BMS Master operates, (c) Condition 3: BMS Slaves operates + BMS Master operates, (d) Condition 4: BMS Master operates + BMS Slaves operates, and (e) Condition 5: BMS Master & Slaves operate simultaneously.

0.15 V, $\Delta V_{b,\max} = 0.55$ V, and $V_{b,\text{diff},\max} = 0.35$ V. BMS Master finishes the battery module balancing process at $t = t_1$ whereas the cell voltages in each battery module are still imbalanced. Condition 3 is the combination of Condition 1 and Condition 2 as shown in Fig. 10(c). The balancing results are the same as Condition 1 till $t = t_1 = 3234$ s. From $t = t_1$ to $t = t_2$, the balancing results are similar to Condition 2. BMS Master finishes the battery module balancing process at $t = t_2 = 9045$ s, and the cell voltages in each battery module are still imbalanced. Where, $\Delta V_{bm,\max} = 0.15$ V, $\Delta V_{b,\max} = 0.14$ V, and $V_{b,\text{diff},\max} = 0.07$ V. Condition 4 is the combination of Condition 2 and Condition 1 as shown in Fig. 10(d). The balancing results are the same as Condition 2 till $t = t_1 =$

8573 s. From $t = t_1$ to $t = t_2$, the balancing results are similar to Condition 1. BMS Slaves converged the cells in the corresponding battery module and finished the balancing process at $t = t_2 = 6075$ s. Where, $\Delta V_{bm,\max} = 0.37$ V, $\Delta V_{b,\max} = 0.11$ V, and $V_{b,\text{diff},\max} = 0.03$ V. For Condition 5, the BMS Master and Slaves work simultaneously and contribute in the balancing process as shown in Fig. 10(e). The balancing process took 9205 s to achieve the balanced condition. Where, $\Delta V_{bm,\max} = 0.15$ V, $\Delta V_{b,\max} = 0.06$ V, and $V_{b,\text{diff},\max} = 0.03$ V. It is obvious from the above results of the proposed BMS mathematical model that the BMS Master and the BMS Slaves can interact together during the balancing process. These simulation results of the proposed balancing model under five

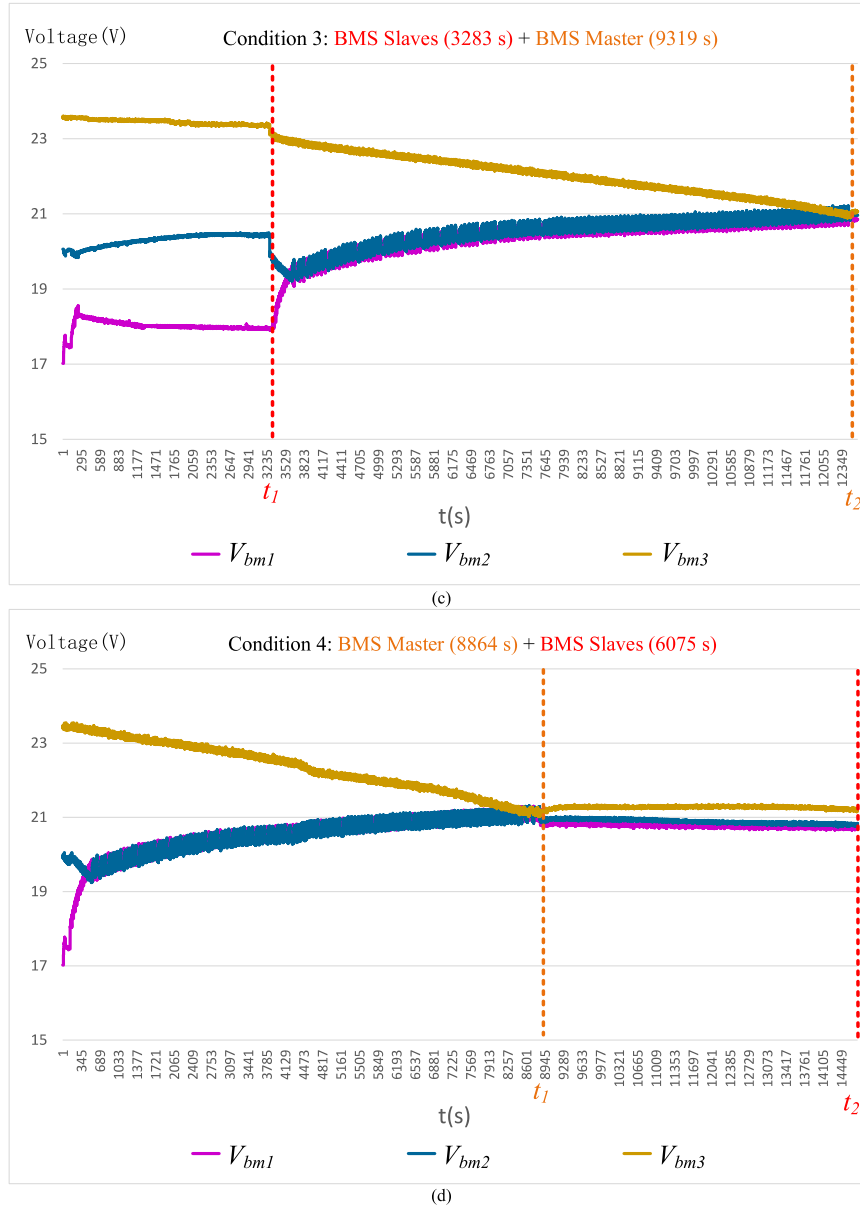


Fig. 13. (continued).

different conditions are concluded and displayed in TABLE 1.

4.2. Hardware implementation

The hardware implementation of the prototype of the proposed MS-BMS architecture has been carried out to validate the proposed balancing model. Fig. 11 shows the prototype of the proposed BMS architecture. Fig. 11(a) displays the hardware of the BMS Slave and Fig. 11(b) displays the hardware of the BMS Master. BMS Master and BMS Slave are combined and connected to the control core to form a complete hardware setup of the proposed MS-BMS which is demonstrated in Fig. 12. Six cells (each having a voltage range of 15 V–25.2 V) are connected in series to form a battery module and the BMS Slaves provide the balancing among the cells of the respective module. The BMS Master performs the balancing operation in the battery pack formed by the connection of three battery modules. The specifications of the components used in the battery module balancing circuit are presented in TABLE 2. To analyze the accurate results of the balancing model and hardware experiment, the corresponding initial voltages of the 18S1P

pack are described in TABLE 3. The experiments have been performed under certain cases as shown in TABLE 4.

4.3. Experimental results

The hardware implementation of the prototype of the proposed MS-BMS architecture has been carried out to validate the proposed balancing model. The measured voltage of each cell under different balancing conditions can be obtained from Fig. 13. To achieve the balanced condition, the process took 3283 s for Condition 1, 8573 s for Condition 2, 12602s (3283 + 9319 s) for Condition 3, 14939s (8864 + 6075 s) for Condition 4 and 9767 s for Condition 5 which are displayed in Fig. 13(a), (b), (c), (d), and (e) respectively. The information about $\Delta V_{bm,max}$, $\Delta V_{b,max}$, and $V_{b,diff,max}$ can be gathered from Fig. 13 which has been mentioned in TABLE 5.

4.4. The validation of the proposed BMS architecture

The hardware implementation of the prototype of the proposed

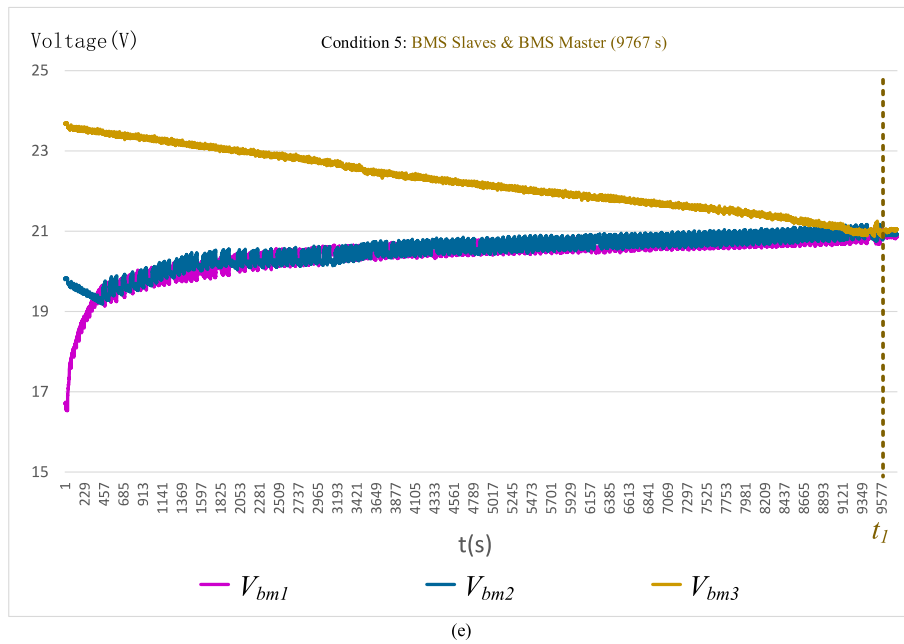


Fig. 13. (continued).

Table 5

The hardware results of the proposed BMS: Balancing performance under five different conditions.

Condition	Step 1 BMS Status	Duration	$\Delta V_{bm,max}$	$\Delta V_{b,max}$	$V_{b, diff,max}$	Step 2 BMS Status	Duration	$\Delta V_{bm,max}$	$\Delta V_{b,max}$	$V_{b, diff,max}$
1	Slaves	3283 s	5.44 V	0.93 V	0.03 V	/	/	/	/	/
2	Master	8573 s	0.17 V	0.60 V	0.45 V	/	/	/	/	/
3	Slaves	3283 s	5.44 V	0.93 V	0.03 V	Master	9319 s	0.21 V	0.08 V	0.03 V
4	Master	8864 s	0.17 V	0.60 V	0.45 V	Slaves	6075 s	0.45 V	0.11 V	0.03 V
5	Master Slaves	9767 s	0.17 V	0.03 V	0.01 V	/	/	/	/	/

Table 6

Comparison of the simulation and hardware results of the proposed BMS under various conditions.

Condition	Simulation Duration	$\Delta V_{bm,max}$	$\Delta V_{b,max}$	$V_{b, diff,max}$	Hardware experiment Duration	$\Delta V_{bm,max}$	$\Delta V_{b,max}$	$V_{b, diff,max}$
1	3234 s	4.61 V	0.82 V	0.03 V	3283 s	5.44 V	0.93 V	0.03 V
2	8573 s	0.15 V	0.55 V	0.35 V	8864 s	0.17 V	0.60 V	0.45 V
3	12,279 s	0.15 V	0.14 V	0.07 V	12,602 s	0.21 V	0.08 V	0.03 V
4	14,648 s	0.37 V	0.11 V	0.03 V	14,939 s	0.45 V	0.11 V	0.03 V
5	9205 s	0.15 V	0.06 V	0.03 V	9767 s	0.17 V	0.03 V	0.01 V

Table 7

Comparisons of the characteristics of certain existing BMS architecture and the proposed BMS.

Ref.	Balancing skill	Balancing circuit power		Equalization speed	Control simplicity	Modularization simplicity	Cost
Proposed	BMS Master + BMS Slaves P to MOD + $\left(\begin{smallmatrix} C \text{ to MOD,} \\ \text{MOD to C} \end{smallmatrix} \right)$	BMS Master 25W	BMS Slaves 10W	H	H	H	M
[26]	C to C	0.176 W~0.812 W		L	L	L	H
[27]	C to P, P to C	3.31 W		M	M	L	L
[28]	$\left(\begin{smallmatrix} \text{Cell} \\ \text{balancing} \end{smallmatrix} \right) + \left(\begin{smallmatrix} \text{Module} \\ \text{balancing} \end{smallmatrix} \right)$	Cell balancing	Module balancing	L	H	H	M
	C to C + P to P	0.9 W		3.6 W			

C–Cell; MOD–Module; P–Pack / H–High; M–Medium; L–Low.

MS–BMS architecture has been carried out to validate the proposed balancing model. Both the simulation and hardware results are discussed in Section 4.1 and 4.3 respectively, which shows the performance and verify the feasibility of the proposed system. The simulation and hardware results are summarized in TABLE 6 to make a comparison

between them. This observation shows the balancing ability and extension possibility of the proposed MS–BMS architecture. It is obvious from the table that the balancing results are influenced and vary with the testing condition. On comparing the experimental results with the simulation results, an absolute error of 1.51%, 3.39%, 2.63%, 1.98%,

and 6.10% is observed in the balancing duration of the proposed balancing model under five different test conditions respectively, whereas the average absolute error is 3.12%.

Condition 1 verifies that the BMS Slave provides the balancing of each cell of the corresponding battery module. However, BMS Slaves cannot contribute to the balancing process among the modules of the battery pack. Similarly, Condition 2 proves that the BMS Master provides the balancing of each module in the battery pack but it cannot contribute to the balancing process among the cells of the battery module. The results of Conditions 3–5 show the significance of the balancing functions' order. If the BMS Slaves operate before the BMS Master as in Condition 3, the voltage differences among cells in each module will be affected during the BMS Master's balancing process. The BMS Slaves have to balance the corresponding battery module again to reduce the difference caused by BMS Master. Conversely, if the BMS Master operates before the BMS Slaves as in Condition 4, the voltage differences among the battery modules will be affected during the BMS Slaves' balancing process. The BMS Master has to balance the battery modules again to reduce the difference caused by BMS Slaves. However, the proposed BMS will have better balancing results and balancing duration if the BMS Master and Slaves work simultaneously as in Condition 5.

4.5. Comparative analysis

The proposed MS-BMS architecture has been implemented, and its performance has been demonstrated through the experimental results. The performance of the proposed BMS has been compared with the other few existing modular BMS architecture [26–28] to show its supremacy. A multi-winding transformer has been used in the BMS [26] for balancing, which restricts the performance of the balancing function. Likewise, the BMS in [27] also employs a multi-winding transformer, which affects its specification expansibility for a high-series battery system. Conversely, the BMS in [28] has better performance than the BMS in [26] and [27]. However, when the voltage between the balancing cells and the voltage difference among balancing modules become closer, the power is lower, and the equalizing speed goes down. The characteristics that are compared are the balancing skills, balancing circuit power, equalization speed, control simplicity, modularization simplicity, and cost. This comparison has been presented in TABLE 7. This comparison demonstrates that the proposed BMS is preferred over the other existing technologies discussed in [26–28] due to its better performance and fine system characteristics. Since there is no direct correlation between the operation of the cell balancing circuit, i.e., the BMS Slaves, and the module balancing circuit, i.e., the BMS Master, the proposed balancing model can be realized as higher power balancing circuits for achieving the optimum BMS functions. If the proposed MS-BMS is applied to real applications, the cells will be divided into 6S1P modules, and each module will be connected in series to meet the requirement specification, such as 48 V, 400 V, etc. Then the module balancing circuit will be designed for the chosen specification.

5. Conclusion

In this study, the MS-BMS balancing model has been proposed. The BMS Slaves provide the balancing of each cell of the battery module, while the BMS Master provides the balancing of each module in the battery pack. Modularized BMS measures the various parameters, and the data are collected by the control core through the Master-Slave Modbus communication protocol and are stored in some data storage system to ensure the safe operation of the proposed system. The proposed balancing model has been simulated in the Simulink model, and the results have been validated on the experimental prototype under five different conditions. On comparing both the results, an absolute error of 1.51%, 3.39%, 2.63%, 1.98%, and 6.10% is observed in the balancing duration of the proposed balancing model under five different test

conditions, respectively, whereas the average absolute error is 3.12%. It is deserved to be mentioned that the proposed balancing model is also applicable and can be used to evaluate the balancing function of other BMS typologies if the equations change to the corresponding characteristics of the BMS architectures. Lastly, the comparison of the proposed BMS with the existing technologies shows the superiority of the proposed BMS in various aspects of system characteristics, including equalization speed, control simplicity, modularization simplicity, and cost. In the future, this study can extend the reliability of the proposed system to hybrid ESS as the electric storage system is not only a BESS these days. Implementing a hybrid ESS with supercapacitors and batteries can meet the need in several applications having high-frequency component loads, such as electric vehicles.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article.

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